

FUNCTIONAL REQUIREMENTS DOCUMENT

Next-Generation Real Estate Investment & Binary Affiliate Management Portal

Document Class: Comprehensive Functional Blueprint (FRD)

Target Domain: Real Estate Asset Sales & Multi-Tier Binary Network Infrastructure

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1. Executive & System Architecture Overview

This project documentation defines the complete operational blueprint for an integrated **Real Estate CRM and Multi-Tier Affiliate Management Portal**. The application serves a dual-purpose environment: enabling investors to secure structural land assets while empowering corporate affiliate networks to distribute properties and generate volume-driven matching commissions through a synchronized hierarchy.

1.1 Unified Core System Architecture

To achieve enterprise scalability, high availability, and rapid transaction security, the core architecture is decoupled into distinct processing layers:

- **Frontend User Interface Panel:** Structured as a Single Page Application (SPA) using a modern utility-first CSS framework. It isolates views strictly by secure routing roles (Public, Authenticated Affiliate, Executive Auditor).
- **Application API Routing Layer:** An asynchronous stateless service controller managing input sanitization, strict JWT-based session boundaries, and structured JSON data contracts.
- **Persistent Storage Layer:** A relational system processing atomic real estate booking contracts, transactional ledgers, and identity tracking schemas with cascading reference constraints.
- **Distributed Cache Engine:** A high-performance in-memory layer specifically managing volatile session states, multi-node tracking matrix caches, and computational lock mechanisms during complex binary network tree refreshes.

1.2 Functional Workflow Boundaries

The system coordinates complex background jobs and data changes without cross-contaminating modules. The operational grid is structured across five major boundaries:

Functional Module	Primary System Responsibility	Critical Performance Constraint
Authentication & Gate	Sponsor node alignment and identity token security.	Real-time upstream validation.
Real Estate Engine	Plot grids, reservation timelines, and structural EMI terms.	Lock-state execution constraints.
Network Matrix Tree	Binary tree placements, spillover logic, downline views.	Recursive layout optimization.
Dual-Wallet Ledgers	Split balance allocations, processing commissions, manual cashouts.	Double-entry accounting checks.

2. Public Gateway & Affiliate Onboarding Module

The public layer governs authentication and onboarding. Because network node integrity is critical to commission tracking, the signup funnel enforces algorithmic link checks before any new account write occurs.

2.1 Sign-In Interface Protocols

The entry portal requires standard secure email/username fields paired with a password hash validator. Security rules include:

- **Persistent Session Storage:** A checkbox permitting token extension inside local client caches, securing user access across sessions.
- **Localization Engine:** A dynamic dropdown switcher supporting multi-language rendering (English, Russian, Hindi). Switching modifies text mapping without modifying underlying data structures.
- **Self-Service Recovery:** A forgotten-password interface triggering verified reset links to registered email profiles with short token expirations.

2.2 Affiliate Onboarding Rules & Tree Alignment

The enrollment workflow establishes a permanent location within the binary organization structure. The layout maps exactly as follows:

Input Parameter	Validation & Business Engine Constraints
Sponsor ID	Mandatory. System runs a background check to confirm the ID exists. If found, it displays the sponsor's name for confirmation. If not found, registration is blocked.
Target Leg Selection	Explicit Option: User selects Left Leg or Right Leg. If the sponsor uses an automated placement rule, the engine queries the database to find the next available position at the bottom of that side.
Security Validation	Enforces unique validation on both Email and Mobile Phone registers. Passwords must pass standard strength checks.

3. Authenticated Affiliate Dashboard UI/UX Elements

Upon successful verification, users are routed to a dashboard providing high-level summaries of their real estate assets and binary networks.

3.1 Interactive Interface Layout

The interface utilizes an active left-sidebar navigation tree paired with a top tracking bar displaying profile states, language overrides, and security controls.

3.2 Core Dashboard Information Components

- **Property Portfolio Metrics:** Displays the total number of plots booked, cumulative payments made, and outstanding liabilities due across all property assets.
- **Financial Earning Trackers:** Live balances summarizing direct sales income, network match commissions, and monthly incentive rewards.
- **Binary Volume Monitors:** Compares Left Leg business volume versus Right Leg business volume. Shows active team point balances to help users spot commission-triggering imbalances.
- **Network Growth Trackers:** Displays cumulative totals for both direct and downline team members across left and right organizational lines.

4. Real Estate & Property Inventory Control Engine

This module manages property inventory, calculates structured monthly payments, and securely houses legal real estate documents.

4.1 Interactive Development Plot Grid Layout

Development projects are rendered visually as an interactive plot matrix grid. Each cell represents an isolated plot. It uses color-coded states to manage inventory allocation transparently:

Status Classification	Visual Cue	System Operational Behavior & Constraints
Available Asset	Green	Open for reservation. Clicking triggers the Booking Wizard and displays cost, size, and EMI options.
On Token Hold	Yellow	An partial token deposit has been verified. The plot is locked for 48 hours. If final booking documents are not completed, it reverts to Green.
Fully Booked / Sold	Red	Fully allocated to an investor. The booking engine disables interactions for other users.

4.2 Dynamic Installment (EMI) Ledger & Document Vault

When an asset is secured, the system generates a tailored payment tracking ledger:

- **Flexible EMI Terms:** Supports standard 12, 24, or 36-month interest structures based on down payment variables.
- **Automated Payment Reminders:** Sends platform notifications and updates the ledger before payment due dates.
- **Secure Document Vault:** Provides direct access to RERA allocation certificates, official plot receipts, and legal land contracts once payment milestones are verified.

5. Affiliate Network Tree & Relationship Visualizer

The organization visualizer renders downline networks in real-time, allowing users to track performance and scale recruitment efficiently.

5.1 Interactive Binary Graphic Interface

The system builds an organization chart starting from the user's root account node. It supports the following interactions:

- **Double-Click Navigation:** Clicking any downline node shifts the focus, making that account the temporary root of the view to simplify deep network tracking.
- **Empty-Node Slot Registration:** Empty slots display a placeholder box. Clicking it opens a registration form with the Sponsor ID and Target Leg prefilled based on that location.
- **Hover Performance Summaries:** Hovering over any node displays a popover card showing their current Rank, total sales volume, and total downline counts.

5.2 Downline Reporting Infrastructure

To manage large teams, users can switch to flat data tables featuring advanced search and filtering tools:

- **Direct Referrals View:** Lists accounts recruited directly by the user, showing their onboarding date, activation status, and total purchase history.
- **Total Network Index:** Displays all accounts within the downline tree. Includes filters for account status (Active, Hold, Pending KYC) and organizational side (Left Leg vs. Right Leg).

6. Dual-Wallet Financial System & Compensation Logic

The application maintains a dual-wallet framework to balance platform liquidity while ensuring property payments remain secure.

6.1 Dual-Wallet Framework Isolation

Every user account tracks two separate balance registries:

Wallet Classification	Fund Input Channels	Operational Restrictions & Rules
Main Cash Wallet	Direct sales commission, binary team matching awards.	Fully withdrawable via bank transfer or usable for property bookings.
Property Installment Wallet	Performance incentives, promotional rewards, optional manual top-ups.	Locked exclusively for land purchases and upcoming EMI payments. Cannot be cashed out directly.

6.2 Systematic Compensation Rule Set

Commissions are calculated automatically using predefined business logic paths:

- **Direct Referral Bonus:** Earned instantly when a direct recruit purchases a property asset. The bonus is based on a flat percentage of the down payment.
- **Binary Team Matching Payouts:** Evaluated on a recurring cycle by comparing the total sales volume of the Left Leg against the Right Leg. It pairs matching volumes at a set ratio (e.g., 1:1), deposits the commission into the Cash Wallet, and carries forward any unpaid volume to the next cycle.
- **Monthly Business Promotion Incentives:** Awarded to affiliates who maintain specified team sales volumes over consecutive months. These rewards route to the Property Wallet to help pay down active property balances.

7. Compliance, Administrative Audits & Core Operations

To meet real estate and financial standards, user activities must pass administrative verification workflows.

7.1 KYC Identity Verification Flow

Before requesting bank transfers or finalizing land deeds, users must submit formal identification. The workflow requires:

- **National Identity Uploads:** Document captures for PAN Card and Aadhaar verification.
- **Settlement Bank Verification:** Account numbers and IFSC codes, cross-checked against uploaded bank documents.
- **Audit Flow States:** Submissions enter a pending queue for administrative review. Financial withdrawals remain locked until the KYC state shifts to 'Approved'.

7.2 Helpdesk Support Ticketing

An internal support system allows authenticated users to open help tickets for payment issues, account errors, or documentation bugs. Tickets track through Open, In-Progress, and Resolved states with complete history logs.

8. Phase-by-Phase Deployment Plan

Rebuilding this platform from scratch follows a structured four-phase timeline to minimize risks and ensure system stability:

Phase	Target Scope	Key Functional Outcomes
Phase 1	Core Framework & Schemas	Establishes database connections, core authentication, and secure session management.
Phase 2	Network Tree & Real Estate Grid	Builds visual organization trees, plot management controls, and real-time inventory updates.
Phase 3	Compensation & Wallets	Deploys calculation engines, tracking ledgers, and automated payment workers.
Phase 4	KYC & Security Hardening	Finalizes identity workflows, access reviews, and high-load performance tests.